SET 7 #14

S_{TRUCK}: $s = ut + \frac{1}{2}at^2$ $\qquad$ $S_{TRAIN} = vt$

$\Rightarrow 117 = 18.0t + \frac{1}{2}at^2$ — (1) $\qquad$ $= 18.0t$ — (2)

Use $v^2 = u^2 + 2as$ to find a.

$\Rightarrow 0 = (18)^2 + 2a(117)$

$\Rightarrow a = -1.385\ ms^{-2}$

$\therefore S_{TRUCK} = 117 = 18.0t + \frac{1}{2}(-1.385)t^2$

$\Rightarrow 0.6923t^2 - 18.0t + 117 = 0.$

To find t:

$$t = \frac{18.0 \pm \sqrt{(-18.0)^2 - 4(0.6923)(117)}}{2(0.6923)}$$

$$= \frac{18.0 \pm 0.06}{1.385}$$

$= 13.04\ s$ or $12.95\ s$. should be equal! (effect of sig. figures)

Sub for $t = 13.04\ s$ in (2) $\qquad$ $t = 12.95\ s$

$\Rightarrow S_{TRAIN} = (18.0)(13.04)$ $\qquad$ $\Rightarrow S_{TRAIN} = (18.0)(12.95)$

$= 234.7\ m$ $\qquad$ $= 233.1\ m$

$\therefore S_{TRAIN} = 233.9\ m$ (average)

$\therefore$ train is 117 m from the truck when it stops.